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GRAPHICS & DESIGN - MECE E3408

Reverse Engineered Toy Plane CAD Design

Design Intent:

For this project we must reverse engineer and model a toy in solidworks. We are to choose the toy from a given selection of toys and demonstrate our abilities to recreate a digital model of the toy. My goal in this project is to demonstrate my ability to break the toy into its constituent parts and recreate the toy without detracting from its original design. Understanding that this is a model of a toy will allow me to make certain simplifications, which I will specify later.

The toy I chose to model is a blue and red single wing propeller plane (*see right*). The plane is made of 7 unique parts: (1) top half of the body, (2) bottom half of the body, (3) engine, (4) propeller, (5) landing gear, (6) wheels, and (7) cockpit. Each part will be modeled then assembled together as an assembly at the end.



Design Simplifications:

(1+2) Plane body (top and bottom half):

- Aesthetic simplifications
 - Exact curvature not taken. Eyeballed curvature by tracing picture
 - Eyeballed thickness and shape of airfoils (including empennage)
 - Logo and words on wings not included
- Structural simplification
 - On the toy there is a small bulge to ensure the two halves fit together. This is not needed for the virtual model

Engine:

- On the inner part of the engine the area around the coils there is a small indent that contributes negligibly to the aesthetics. It is therefore excluded from the model

Propeller:

- Exact curvature and shape not maintained, but closely approximated.

Landing Gear:

- The overall shape is not 1:1. The ‘character’ of the landing gear was maintained using two curved legs connected across the middle. Exact curvature not maintained.
- The shape of the piece that slots into the body of the plane is rectangular to simplify design.
- The ‘axles’ shape is designed to hold the wheel virtually, but is not designed with tolerance to allow the wheel to spin in a real world environment

Wheels:

- Small aesthetic ring excluded

Cockpit:

- Top cross section is an ellipse to simplify design. Does not detract aesthetically.
- Overall taking exact dimensions for the cockpit was difficult so the goal was to maintain the general shape and spacing of the sub-parts.
- Small bumps on the cushions are not modeled since the overall shape already clearly indicates that the seat is a seat.

Modeling Plan:

(1+2) Plane body (top and bottom half): The main part of the toy is the body of the plane which includes the wings, tail, and fuselage. Since the top and bottom half come together to form a continuous curved body, they will be modeled together initially as a single part then split at when completed. Here it is important to maintain the general shape of the plane so it still looks like a plane, but since this is not a real plane it is okay if the curved surfaces are not exactly the same as the toy and it is okay if the wings are not airfoils that can create lift. Additionally the words on the plane are not necessary and will not be modeled.

1. Create the fuselage using a base revolve.
2. Create wing 1 using loft or similar tool
3. Mirror the wing over the center of the fuselage

4. Create one of the elevators using same method as the wings
5. Mirror elevator to other side of the plane
6. Create vertical stabilizer using same method
7. Add back 'landing gear,' also by lofting downward
8. Insert reference plane on an angle to create the shape behind the cockpit. This will be modeled as a circular loft into the toy.
9. Create another reference plane to sketch a semicircle and revolve cut the indent out of the area behind the cockpit
10. Extrude cut ellipse from top of the plane to make room for the cockpit. This is not the exact shape but it is close and makes modeling the rest of the cockpit simpler while maintaining the general appearance.
11. Extrude cut fillet rectangle out of the bottom of the plane. Again this is not the same shape as the cutout for the landing gear on the toy, but it is close and can serve the same purpose.
12. Extrude circle on the front of the toy for the connection to the engine
13. Split the top and bottom

Engine:

1. Base revolve main part of the engine
2. Extrude cut circle connection to the rest of the plane
3. Extrude center of engine
4. Create on the the coils inside of the engineer
5. Circular pattern the coil around the center of the engine

Landing Gear: The exact geometry of the landing gear is not crucial, but to maintain appearance curves will be similar to that of the toy. Some simplifications may need to be taken to ease the creation of the thin lofts.

1. Copy the bottom of the plane and remove everything but the central revolve
2. Cut outside of the area cut for the landing gear to slot in. this is to maintain the curvature on the landing gear piece
3. Sketch out top and bottom parts of the landing gear then loft
4. Cut out the semi circle (can be viewed from in front of the plane)
5. Add circles/axles to later hold the wheels

6. Add cross beam between the two wheels

Propeller:

1. Revolve to create center of the propeller
2. Create one blade of the propeller using lofts
3. Circular pattern to create the three blades

Wheels:

1. Revolve wheel cross section
2. Add ident
3. Circular pattern

Cockpit: Goal is to make the sense that it is a cockpit, so some details can be glossed over, but there still needs to be some to give the feel of a cockpit.

1. Extrude wall of the cockpit
2. Extrude seat
3. Extrude seat cushion
4. Create reference plane for the dashboard
5. Extrude up to the plane for the dashboard
6. Extrude the joystick
7. Loft to create the windshield
8. Add details

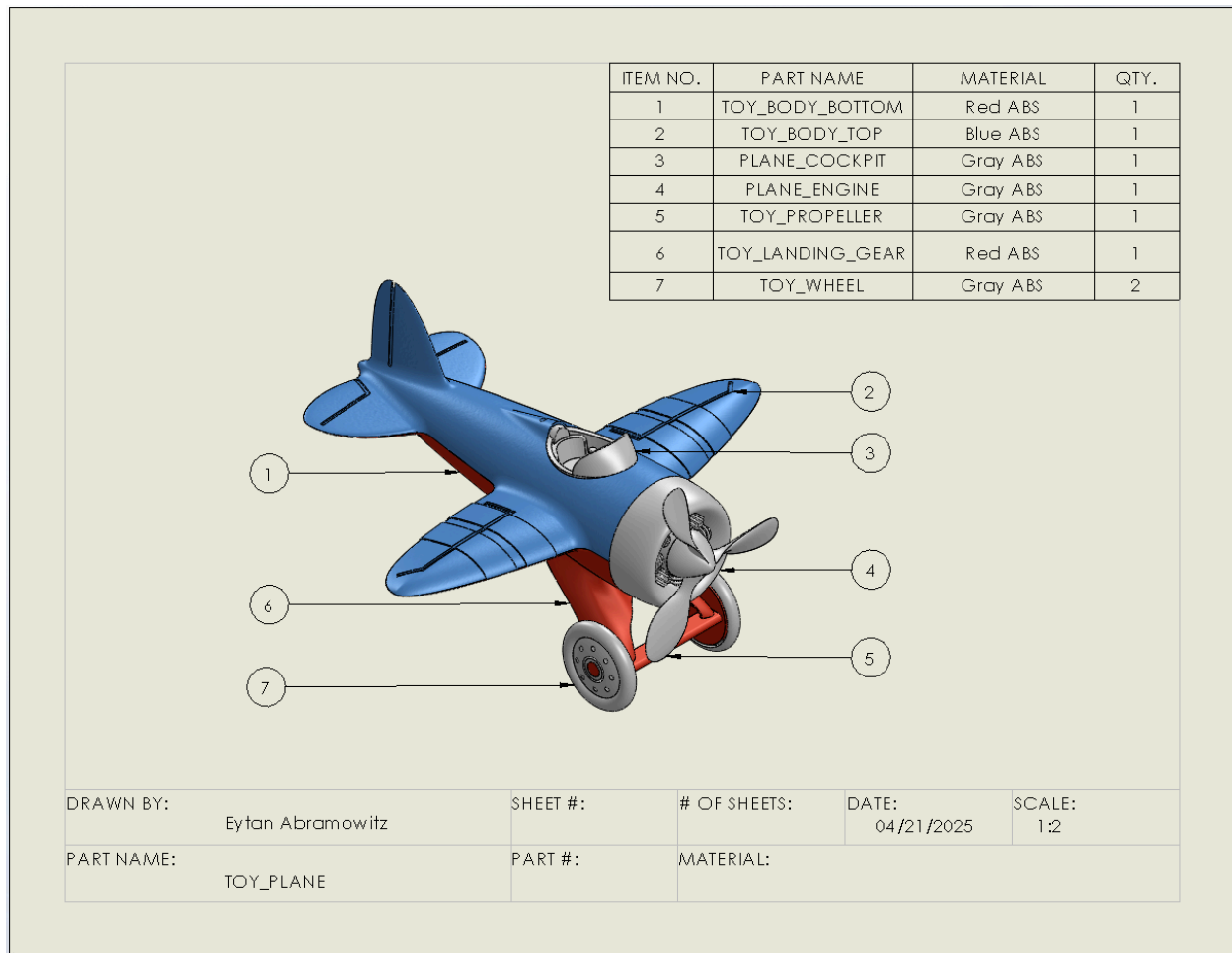
Difficulties Encountered While Modeling:

Initially I expected the most difficult part of the modeling process to be modeling the wings and curved surfaces throughout the plane. While this was difficult, once I figured out how to properly draw 3D sketches using splines and how to use the 'Boundary Boss/Base' tool I was able to apply similar steps to creating all the different wing shapes on the plane, including the propeller. I was even able to use the same tool to create the landing gear. While the creation of landing gear legs sometimes failed, with proper tweaking they generated properly.

Additionally, I was unsure how to model the surface features on the wings, but after some research I learned how to use the wrap tool and offset surface tool to create the features on the wings.

Unexpectedly, the geometry of the cockpit was the hardest part to model. While it appears that the cockpit is made mostly of relatively simple geometry, the order of operations was critical in order to create the correct final product. While modeling I needed to change the order a couple times to allow for extrudes (cuts/bases) to interact with some parts but not others. I also needed to make small adjustments on cut geometry since there were a couple spots I thought it would cut through, but it did not cut everything.

Assembly Drawing:



Final Results

Original Toy:



Solidworks Model

